

Engine(s) Have No or Only Partial Response to Throttle or Cummins® Inboard Joystick Operation

Symptoms

- Throttle will **not** activate or partially function when used.

How To Use This Tree

This symptom tree can be used to troubleshoot a malfunction. Start by performing Step 1 troubleshooting. Step 2 will ask a series of questions and will provide a list of troubleshooting steps to perform, depending on the symptom.

Shoptalk

- Possible causes include:
- Open signal circuit in throttle Signal
 - Malfunctioning throttle
 - Malfunctioning backup throttle
 - Inboard joystick is malfunctioning.
- Reference the appropriate original equipment manufacturer (OEM) wiring diagram when troubleshooting circuits that utilize wiring supplied by the OEM.

Troubleshooting Summary

STEPS		SPECIFICATIONS
STEP 1.	Check vessel for fault or alarm codes.	
	STEP 1A. Check vessel for Cummins® inboard joystick or Cummins® Throttle Shift.	Cummins® inboard joystick or Electronic Throttle and Shift installed?
	STEP 1B. Read the alarm codes.	Alarm code indicator light illuminated?
	STEP 1C. Read the engine control module (ECM) fault codes.	Throttle or sensor supply faults active?
STEP 2.	Check the throttle circuit.	

STEPS		SPECIFICATIONS
	STEP 2A. Check the engine interface wiring harness throttle voltage.	Between 4.75 VDC and 5.25 VDC?
	STEP 2B. Check the engine interface wiring harness throttle SIGNAL.	Less than 10 ohms?

STEP 1. Check vessel for fault or alarm codes.

STEP 1A. Check vessel for Cummins® inboard joystick or Cummins® Throttle Shift.

Conditions :

- Turn System Enable switch OFF.

Action	Specification/Repair	Next Step
Check vessel for Cummins® inboard joystick or Cummins® Throttle Shift	Cummins® inboard joystick or Electronic Throttle and Shift installed? YES	1B
	Cummins® inboard joystick or Electronic Throttle and Shift installed? NO	1C

STEP 1B. Read the alarm codes.

Conditions :

- Turn System Enable switch ON.

Action	Specification/Repair	Next Step
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Read the alarm codes. Record the sequence of alternating slow and fast blinking indicator lights.	Alarm code indicator light illuminated? YES Repair : See the Cummins® Electronic Throttle and Shift and Cummins® Inboard Joystick Marine Controls MC101 Master Repair Manual.	Go to the appropriate alarm code troubleshooting tree
	Alarm code indicator light illuminated? NO	1C

STEP 1C. Read the ECM fault codes.

Conditions :

- Turn System Enable switch ON.
- Connect INSITE™ electronic service tool.

Action	Specification/Repair	Next Step
Read the ECM fault codes. Use INSITE™ electronic service tool to read the fault codes.	Any throttle or sensor supply faults active? YES Repair : See the Cummins® fault code manual for the engine.	Go to the appropriate fault code troubleshooting tree.
	Any throttle or sensor supply faults active? NO	2A

STEP 2. Check the throttle circuit.

STEP 2A. Check the engine interface wiring harness throttle voltage.

Conditions :

- Turn System Enable switch ON.
- Disconnect the throttle harness from the 3 pin connector on the engine interface wiring harness.

Action	Specification/Repair	Next Step
Check the engine interface wiring harness throttle voltage. • Place one test lead on Pin A of the engine interface wiring harness 3 pin throttle connector. • Place the other lead on Pin C of the engine interface wiring harness 3 pin throttle connector.	Between 4.75 VDC and 5.25 VDC? YES	2B
	Between 4.75 VDC and 5.25 VDC? NO Repair : An open Supply or Return is detected in the engine interface harness or engine wiring harness. Troubleshoot and repair the harness. • Engine interface harness: Refer to Procedure 015-093 in Section 15 (/qs3/pubsys2/xml/en/procedures/513/513-015-093.html). • Engine wiring harness: See Procedure 019-043 in the Cummins® service manual for the engine.	Repair complete.

STEP 2B. Check the engine interface wiring harness throttle SIGNAL.**Conditions :**

- Turn System Enable switch OFF.
- Disconnect the throttle harness from the 3 pin connector on the engine interface wiring harness.
- Disconnect the harness from the throttle or throttle control processor module

Action	Specification/Repair	Next Step
Check the engine interface wiring harness throttle SIGNAL. Measure the resistance of the throttle SIGNAL pin at the engine interface harness to the throttle SIGNAL pin at the throttle or the throttle control processor module.	Less than 10 ohms? YES Repair : - Refer to the appropriate Symptom in the Cummins® Electronic Throttle and Shift and Cummins® Inboard Joystick Marine Controls MC101 Master Repair Manual. - For Non Cummins® throttle refer to throttle OEM service manual	Go to the appropriate alarm code troubleshooting tree.
	Less than 10 ohms? NO Repair : An open throttle signal has been detected. Repair or replace the harness or interconnects.	Repair complete.